# musicar pretrained bases

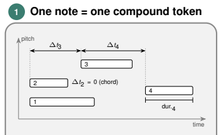
Three decoder-only transformers over compound note tokens (one position = one note, 20 ms grid), pretrained without captions. Each diagram has its own key.

## musicar 450M base (464M)

Pretrained multi-instrument MIDI model without text captions: 464.24M parameters (trunk 445.98M, cascade heads 16.28M, embeddings 1.99M), trained on 8.3B notes.

How to read: numbers give the reading order; teal = the model's structure; grey boxes = linear layers and embedding tables; dashed frame = a module drawn open in place; dotted frame or box = a zoom into the dotted part.

### 1 One note = one compound token

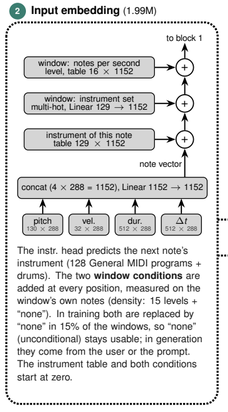



Notes 1–4 become positions 1–4: one position per note, in onset order, each with all five values:

| field | values | meaning |
|---|---|---|
| $\Delta t$ | 512 | time since the previous note started (20 ms steps) |
| **instr.** | 129 | General MIDI program 0–127 or drums |
| **pitch** | 130 | MIDI pitch 0–127 + start / end of a piece (BOS / EOS) |
| **dur.** | 512 | note length (20 ms steps) |
| **vel.** | 32 | velocity: how hard the note is played |

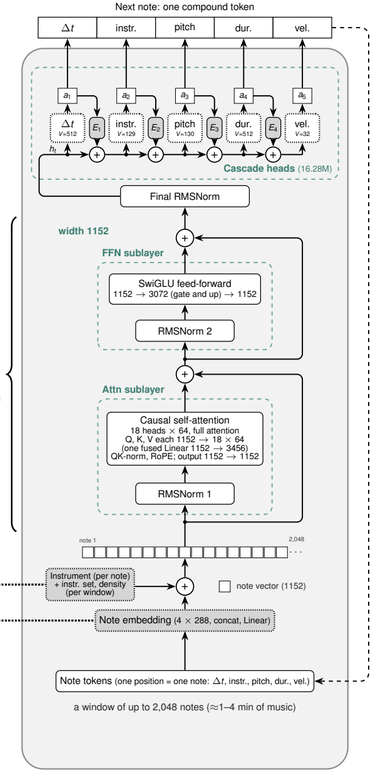
**Time is absolute:** no bars, beats or tempo tokens; $\Delta t$ and durations stop at 10.22 s. The notes of all tracks are merged into one sequence; for drums the pitch is the drum sound.

### 2 Input embedding (1.99M)




The instr. head predicts the next note's instrument (128 General MIDI programs + drums). The two **window conditions** are added at every position, measured on the window's own notes (density: 15 levels + "none"). In training both are replaced by "none" in 15% of the windows, so "none" (unconditional) stays usable; in generation they come from the user or the prompt. The instrument table and both conditions start at zero.

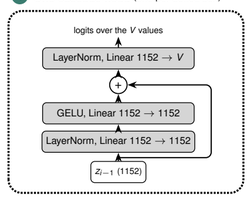
### 3 Music model




28 × (445.98M)

a window of up to 2,048 notes (≈1–4 min of music)

### 4 One cascade head (one per dotted box)




**Why a cascade.** One position predicts all five values of the next note. Head $i$ reads the running state

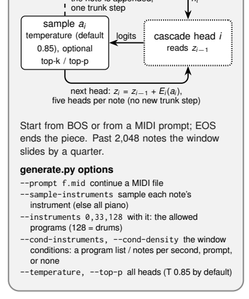
$z_0 = h_t, \quad z_i = z_{i-1} + E_i(a_i),$

the trunk state $h_t$ plus the embeddings $E$ of the values already chosen (tables 512, 129, 130, 512×1152; the last head needs none). So the note's probability follows the chain rule:

$p(\Delta t)\, p(\text{instr.} \mid \Delta t)\, p(\text{pitch} \mid \Delta t, \text{instr.}) \cdots$

Each head has its own weights. $a_i$ is the true value in training (teacher forcing) and the sampled value in generation. Loss: the sum of the five cross-entropies. In a 42M test, the same heads without the chain were +0.34 nats / note worse.

### 5 Generation



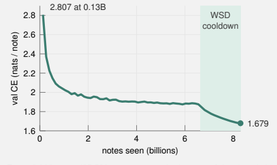

Start from BOS or from a MIDI prompt; EOS ends the piece. Past 2,048 notes the window slides by a quarter.

**generate.py options**

- `--prompt f.mid` continue a MIDI file
- `--sample-instruments` sample each note's instrument (else all piano)
- `--instruments 0,33,128` with it: the allowed programs (128 = drums)
- `--cond-instruments, --cond-density` the window conditions: a program list / notes per second, prompt, or none
- `--temperature, --top-p` all heads (T 0.85 by default)

### 6 Pretraining

**Clean GigaMIDI validation (8.3B notes)**




Every 2,000 steps (65,536 notes per step): the sum of the $\Delta t$, pitch, duration and velocity cross-entropies on GigaMIDI validation files whose notes do not occur in its training split. Constant LR, then the $1 - \sqrt{\cdot}$ cooldown in the last 20%. One run.

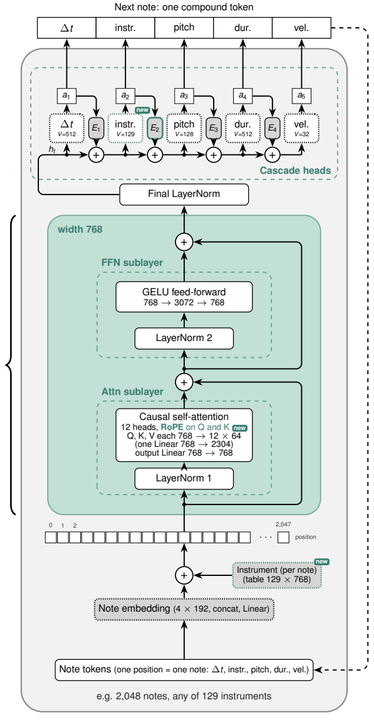
| Setting | Base pretraining (done) |
|---|---|
| Data | GigaMIDI 0.4, Aria-MIDI 0.3, Discover MIDI 0.3 (sampling weights); no audio-derived songs |
| Notes seen | 8.3B (≈1.8 epochs on average; Aria-MIDI ≈4.5) |
| Steps, batch | 126,582 steps × 65,536 notes (32 windows of 2,048 notes) |
| Optimizer | Muon (Moonlight scaling) for the weight matrices of the blocks, AdamW for the rest; one LR $10^{-3}$, weight decay 0.1 |
| Schedule | WSD: 1,000 warmup steps, constant LR, then $1 - \sqrt{\cdot}$ decay to 0 over the last 20% |
| Loss | sum of the five cross-entropies, teacher forcing; no dropout, no label smoothing |
| Augmentation | random transposition by up to ±5 semitones (drums unchanged) |
| Conditions | instrument set and density replaced by "none" in 15% of the windows |
| Precision | bf16 with fp8 matmuls, torch.compile; the checkpoint holds bf16 weights |
| Hardware | one rented RTX 5090, ≈41 h |
| Result | clean GigaMIDI validation **1.679** nats / note = $\Delta t$ 0.170 + pitch 0.367 + dur. 0.644 + vel. 0.497 (the instrument head, 0.117, apart); 1.681 with both conditions "none" |

## musicar 107M band (108M)

107.93M parameters: trunk 99.11M, cascade heads 7.90M, embeddings 0.92M. 129 instruments, context 2,048 notes.

How to read: numbers give the reading order; teal = the repeated block; grey boxes = linear layers and embedding tables; dashed frame = a module drawn open in place; dotted frame or box = a zoom (white panel) into the dotted part; new and teal outlines = added when the piano model was grown (section 4).

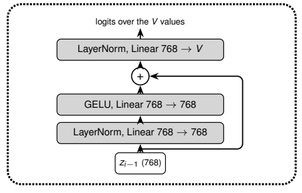
### 1 Music model




14 ×

e.g. 2,048 notes, any of 129 instruments

### 3 One cascade head




**Why a cascade.** Head $i$ (a dotted box, own weights) reads the trunk state plus the embeddings $E$ of the values already chosen,

$z_0 = h_t, \quad z_i = z_{i-1} + E_i(a_i),$

so the note's probability follows the chain rule:
$p(\Delta t)\, p(\text{instr.} \mid \Delta t) \cdots$. In training $a_i$ is the true value (teacher forcing), in generation the sampled one. Loss: the sum of the five cross-entropies.

### 4 Grown from musicar 107M piano

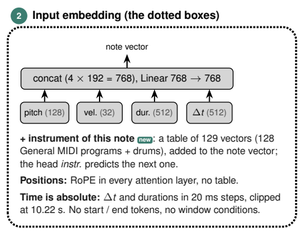
**Copied** (and trained on): the trunk, the note embedding, the four note heads and their condition embeddings.

**New:** the instrument embedding, the instrument head and its condition embedding $E_2$; both tables and the head's output layer start at zero: at step 0 they change nothing.

**Changed:** learned positions → RoPE, context 512 → 2,048 notes (not function-preserving).

**Run:** GigaMIDI 0.7 + Aria-MIDI 0.3, 197M notes (3,000 steps × 65,536), 83 min. GigaMIDI validation 2.232 nats / note; the same recipe from scratch: 2.597.

### 2 Input embedding (the dotted boxes)




**+ instrument of this note** new: a table of 129 vectors (128 General MIDI programs + drums), added to the note vector; the head *instr.* predicts the next one.

**Positions:** RoPE in every attention layer, no table.

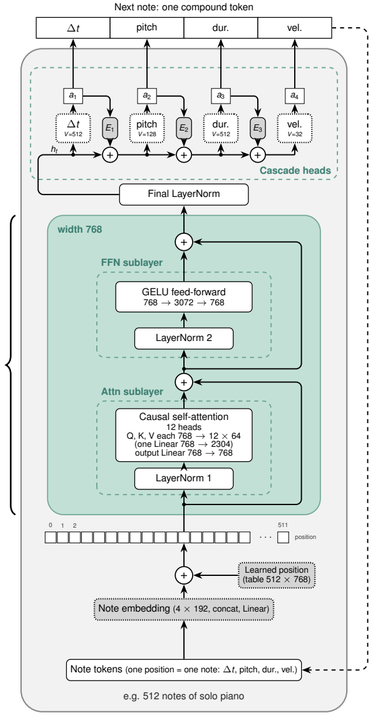
**Time is absolute:** $\Delta t$ and durations in 20 ms steps, clipped at 10.22 s. No start / end tokens, no window conditions.

## musicar 107M piano (107M)

106.84M parameters: trunk 99.11M, cascade heads 6.52M, embeddings 1.21M. Solo piano, context 512 notes.

How to read: numbers give the reading order; teal = the repeated block; grey boxes = linear layers and embedding tables; dashed frame = a module drawn open in place; dotted frame or box = a zoom (white panel) into the dotted part.

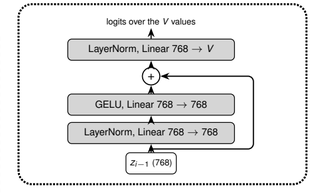
### 1 Music model




14 ×

e.g. 512 notes of solo piano

### 3 One cascade head




**Why a cascade.** Head $i$ (a dotted box, own weights) reads the trunk state plus the embeddings $E$ of the values already chosen,

$z_0 = h_t, \quad z_i = z_{i-1} + E_i(a_i),$

so the note's probability follows the chain rule:
$p(\Delta t)\, p(\text{pitch} \mid \Delta t) \cdots$. In training $a_i$ is the true value (teacher forcing), in generation the sampled one. Loss: the sum of the four cross-entropies.

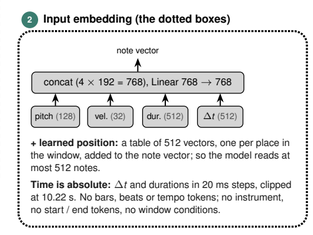
### 4 Training (done, 2026-09-30)

**Data:** MAESTRO + GiantMIDI-Piano (≈5% of the notes) and Aria-MIDI (≈95%), solo piano.

**Run:** 38,000 steps × 61,440 notes (120 windows of 512) = 2.33B notes, ≈4 passes; Muon + AdamW, LR $10^{-3}$, WSD schedule; 12 h 27 min on a laptop RTX 5060 (8 GB).

**Result:** Aria-MIDI validation 5.706 nats / note, inside the 5.47–5.72 forecast of the scaling ladder; MAESTRO + GiantMIDI-Piano validation 6.987.

### 2 Input embedding (the dotted boxes)




**+ learned position:** a table of 512 vectors, one per place in the window, added to the note vector; so the model reads at most 512 notes.

**Time is absolute:** $\Delta t$ and durations in 20 ms steps, clipped at 10.22 s. No bars, beats or tempo tokens; no instrument, no start / end tokens, no window conditions.

## Comparison

| | musicar 450M base | musicar 107M band | musicar 107M piano |
|---|---|---|---|
| Parameters | 464.24M: trunk 445.98M, cascade heads 16.28M, embeddings 1.99M | 107.93M: trunk 99.11M, cascade heads 7.90M, embeddings 0.92M | 106.84M: trunk 99.11M, cascade heads 6.52M, embeddings 1.21M |
| Trunk | 28 blocks, width 1152, 18 attention heads | 14 blocks, width 768, 12 attention heads | 14 blocks, width 768, 12 attention heads |
| Context | 2,048 notes | 2,048 notes | 512 notes |
| Instruments | 129 (128 General MIDI programs + drums), BOS / EOS, window conditions (instrument set, density) | 129 (128 General MIDI programs + drums); no BOS / EOS, no conditions | piano only; no BOS / EOS, no conditions |
| Positions | RoPE | RoPE (replaced the piano model's learned table) | learned table (512 × 768) |
| Norm / MLP | RMSNorm, SwiGLU, QK-norm | LayerNorm, GELU | LayerNorm, GELU |
| Data | GigaMIDI 0.4, Aria-MIDI 0.3, Discover MIDI 0.3 | grown from the 107M piano, then GigaMIDI 0.7, Aria-MIDI 0.3 | MAESTRO + GiantMIDI-Piano ≈5%, Aria-MIDI ≈95% |
| Notes seen | 8.3B (126,582 steps, ≈41 h on one RTX 5090) | 197M (3,000 steps, 83 min on a laptop RTX 5060), after the piano model's 2.33B | 2.33B (38,000 steps, 12 h 27 min on a laptop RTX 5060) |
| Result | clean GigaMIDI validation **1.679** | GigaMIDI validation **2.232** (older split that shares files with training) | Aria-MIDI validation **5.706** (512-note windows) |

Result: validation cross-entropy in nats per note (teacher-forced, summed over $\Delta t$, pitch, duration and velocity; lower is better). The three validation sets differ, so the numbers are not comparable across columns.

## Next (planned, not done)

